

Ans: D

Q27

X-ray spectra are created by electrons impinging on a metal.

Loss in KE of electrons = Energy of X-rays

Maximum KE of electrons - 0 = Maximum energy of X-rays

$$\text{Maximum KE of electrons} = E = \frac{hc}{\lambda_{\min}} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{35 \times 10^{-12}} = 5.68 \times 10^{-15} \text{ J}$$

Ans: C

Q28